

REASONING BASED DATA SUFFICIENCY 1

Directions for questions 1 to 7: This question is followed by two statements, A and B. Study whether the information given in the statements is sufficient to answer the question and then select the correct alternative from the given choices.

Each of the three persons A, B and C belongs to a different category among Truth tellers (who always speak the truth), Liars (who always lie) and Alternators (who make true and false statement alternately). Who among them is the Truth teller?

I. A said, "I am a Truth Teller. B is a Truth Teller".

II. B said, " I am a liar. A is a liar".

- ☒ a) The question can be answered by any one of the two statements alone but not by the other statement alone. (Correct Answer - ✓)
- ☐ b) The question can be answered by either of the two statements alone.
- ☐ c) The question can be answered only if both the statements are taken together.
- ☐ d) The question cannot be answered even if both the statements are taken together.

Solution:

Given, each of A, B and C belongs to a different category among Truth tellers, Liars and Alternators. From statement I it can be said that A is not a truth teller. Either B or C is the truth teller. Thus, I alone is not sufficient.

From II, B is an alternator and his first statement is false and the second one is true. This implies A is the liar and C is the truth teller. Hence, II alone is sufficient.

Choice (A)

Directions for questions 1 to 7: This question is followed by two statements, A and B. Study whether the information given in the statements is sufficient to answer the question and then select the correct alternative from the given choices.

How many among a group of 20 persons, like both Cricket and Hockey?

I. Among them, 15 persons like at least one among Cricket and Hockey.

II. Among them, ten persons like Cricket and eight persons like Hockey.

- ☐ a) The question can be answered by any one of the two statements alone but not by the other statement alone.
- ☐ b) The question can be answered by either of the two statements alone.
- ☒ c) The question can be answered only if both the statements are taken together. (Correct Answer - ✓)
- ☐ d) The question cannot be answered even if both the statements are taken together.

Solution:

From I we can say that five persons like neither Cricket nor Hockey and Exactly one + Exactly two = 15. But this information is not sufficient to answer the question.
From II, Exactly one + 2 x Exactly 2 = 10 + 8 = 18. II alone is not sufficient to answer the question.
By combining both, we can find the number of persons who like both Cricket and Hockey. Thus both the statements are required to answer the question.

Choice (C)

Five persons P, Q, R, S and T are standing in a row facing north, not necessarily in the same order. Is R standing in the middle of the row?

I. S is two places away to the right of R.

II. P is three places away from Q.

- ☐ a) The question can be answered by any one of the two statements alone but not by the other statement alone.
- ☐ b) The question can be answered by either of the two statements alone.
- ☐ c) The question can be answered only if both the statements are taken together.
- ☒ d) The question cannot be answered even if both the statements are taken together. (Correct Answer - ✓)

Solution:

From I, we know that one person stands between S and R. R may or may not be in the middle. Thus I is not sufficient to answer the question.
From II, two persons are between P and Q. Hence, neither P nor Q is in the middle. II is not sufficient to answer the question.
By combining I and II, either S or R is in the middle. Hence, the question cannot be answered even by using information in both the statements.

Choice (D)

Among A, B, C, D, E, F and G, who are sitting in a row, is D at the right most position?

I. The number of persons to the right of A is the same as the number of persons to the left of D. C is at one of the ends.

II. If B and D interchange their positions, each of them gets exactly two new neighbours.

- ☐ a) The question can be answered by any one of the two statements alone but not by the other statement alone.
- ☒ b) The question can be answered by either of the two statements alone. (Correct Answer - ✓)
- ☐ c) The question can be answered only if both the statements are taken together.
- ☐ d) The question cannot be answered even if both the statements are taken together.

Solution:

From I, A and D can be in the positions (1,7) or (2,6) or (3,5). But C is at an end. Hence, A and D cannot be in positions (1,7). Thus I alone is sufficient
From II, B and D are in (2,5) or (2,6) or (3,6). Thus D cannot be at the right most position. Thus II alone is sufficient to answer the question
Hence, each statement alone is sufficient to answer the question.

Choice (B)

A cube is painted in such a way that a pair of faces are in green, a second pair of faces are in blue and the remaining faces are in red. Are the faces painted in blue opposite each other?

- I. The faces painted in green are opposite each other and also the faces painted in red are opposite each other.
- II. The faces painted in green are adjacent to each other and also the faces painted in red are adjacent to each other.

- ☐ a) The question can be answered by any one of the two statements alone but not by the other statement alone. (Correct Answer - ✓)
- ☐ b) The question can be answered by either of the two statements alone.
- ☐ c) The question can be answered only if both the statements are taken together.
- ☐ d) The question cannot be answered even if both the statements are taken together.

Solution:

From I, when green is on opposite faces and red on opposite faces, the remaining faces will be opposite each other. Hence, blue will be on opposite faces. Thus I alone is sufficient.

From II, when green is on adjacent faces and red on adjacent faces, the remaining faces can be opposite or adjacent each other. Hence, II alone is not sufficient. Thus I alone is sufficient to answer the question. Choice (A)

Who among P, Q, R, S and T is the youngest?

- I. The number of persons older than P is the same as the number of persons younger than R.
- II. At least three persons are older than T, who is not older than R.

- ☐ a) The question can be answered by any one of the two statements alone but not by the other statement alone.
- ☐ b) The question can be answered by either of the two statements alone.
- ☐ c) The question can be answered only if both the statements are taken together.
- ☐ d) The question cannot be answered even if both the statements are taken together. (Correct Answer - ✓)

Solution:

I alone is not sufficient to answer the question. From II, P, Q and S are older than T. R and T can be of the same age or T could be younger than R, as it is not mentioned that these five persons are of different ages.
By combining I and II also the question cannot be answered. Choice (D)

It is raining. Did Ramu use umbrella?

I Ramu uses umbrella, only when it rains.

II Ramu doesn't use umbrella, unless it rains.

- ☐ a) The question can be answered by any one of the two statements alone but not by the other statement alone.
- ☐ b) The question can be answered by either of the two statements alone.
- ☐ c) The question can be answered only if both the statements are taken together.
- ☒ d) The question cannot be answered even if both the statements are taken together. (Correct Answer - ✓)

Solution:

From I, when it does not rain it can say that Ramu doesn't use umbrella. But cannot say anything when it rains. Hence, I alone is not sufficient.
From II, rain is a necessary condition for Ramu to use umbrella but not a sufficient condition. Hence, II alone is not sufficient.
By combining both the statements also the question cannot be answered. Choice (D)

Directions for questions 8 to 12: A word and number arrangement machine when given an input line of word numbers rearranges them following a particular rule in each step. The following is an illustration of an input and rearrangement.

Input : 25 water tea cold 84 56 41 hot

Step I : cold water tea 84 56 41 hot 25

Step II : hot cold water tea 84 56 25 41

Step III : tea hot cold water 84 25 41 56

Step IV : water tea hot cold 25 41 56 84

and step IV is the last step of the rearrangement of the above input.

As per the rules followed in the above steps, answer the questions for the given input.

Input: world 62 17 human 54 good book 75.

Which of the following would be step III?

- ☐ a) human good book 75 world 17 54 62
- ☐ b) human good book world 17 75 54 62
- ☐ c) human good book world 75 17 54 62 (Correct Answer - ✓)
- ☐ d) There is no such step

Solution:

By observing the given input and output we can say that, the numbers are arranged in the ascending order and the words are arranged in the reverse alphabetical order. From one step to the next step two operations have taken place. The smallest number and the first word in the alphabetical order are shifted to either ends. The number is shifted to the right end and the word is shifted to the left end.

Input: world 62 17 human 54 good book 75
Step I: book world 62 human 54 good 75 17
Step II: good book world 62 human 75 17 54
Step III: human good book world 75 17 54 62
Step IV: world human good book 17 54 62 75
Step IV is the last step for the given input.

Choice (C) is the step III of the given input.

Choice (C)

Which word/number would be at fifth position from the right end in step II of the input?

- ☐ a) 62 (Correct Answer - ✓)
- ☐ b) human
- ☐ c) 75
- ☐ d) None of these

Which Step would the following output be?

good book world 62 human 75 17 54

- ☒ a) Step II (Correct Answer - ✓)
- ☐ b) Step III
- ☐ c) Step IV
- ☐ d) There is no such step

Which word/number would be at the third position from the left end in step IV of the input?

- ☒ a) good (Correct Answer - ✓)
- ☐ b) human
- ☐ c) 54
- ☐ d) None of these

Which of the following would be step V?

- ☐ a) world human good book 17 62 54 75
- ☐ b) world human good 17 book 54 62 75
- ☐ c) world human good book 17 54 62 75
- ☒ d) No such step exists (Correct Answer - ✓)

Directions for questions 13 to 15: Select the correct alternative from the given choices.

Pointing at a photograph, Mr. Kiran told Mr. Rajesh, "He is the father-in-law of your mother's husband and is my paternal grandfather", how is Mr. Kiran related to Mr. Rajesh?

- ☐ a) Uncle
- ☐ b) Father
- ☐ c) Brother

☒ d) Cousin (Your answer - ✓)

Directions for questions 13 to 15: Select the correct alternative from the given choices.

Four of the following five are alike in a certain way and so form a group. Find the one that does not belong to that group.

☒ a) 91 (Your answer - ✗)

☐ b) 49

☐ c) 51 (Correct Answer - ✓)

☐ d) 77

Solution:

Except 51, the other numbers are divisible by 7.

Choice (C)

Directions for questions 13 to 15: Select the correct alternative from the given choices.

A is 10 m towards to the North of B, which is 6 m to the West of D. B is 12 m towards the East of C. E is 20 m towards the South of C. In which direction is E with respect to A?

☐ a) North-east

☒ b) South-west (Your answer - ✓)

☐ c) South

☐ d) North-west